Lab 3: Architectural Pattern Selection & Justification

Freelance Content Creator Escrow Platform

Name: Bhuvigna Reddy A T SRN: PES2UG24CS119

Architecture Selection

We chose Microservices Architecture for the Freelance Content Creator Escrow Platform.

Architectural Choice

The system is decomposed into independently deployable services — **Project & Proposal Service**, **Contract Service**, **Escrow & Payment Service**, **Notification Service**, **User & Auth Service**, and a shared encrypted **Data Store** — all exposed through an **API Gateway**. This mirrors the functional boundaries already defined in the FR/NFR table (FR-001 to FR-005) and keeps each capability independently testable, deployable, and scalable.

Two Reasons

1. Independent scaling for the performance-critical path: NFR-002 requires user actions and page loads to complete within 3 seconds under normal operating conditions. Because Project & Proposal Service (FR-001) is its own service, it can be scaled horizontally during peak posting and browsing periods without over-provisioning less-used services such as Notification or User & Auth.

2. Fault isolation across independent workflows: Project creation (FR-002), proposal browsing (FR-001), contract creation (FR-003), deliverable submission/approval (FR-004), and escrow release (FR-005) are logically separate workflows that do not need to fail together. If the Notification service degrades, clients and freelancers can still browse projects, submit proposals, and manage contracts — a monolithic layered design would risk a single failure cascading across all features, which is especially risky for a platform holding client funds in escrow.

Security Advantage

NFR-001 requires that user data, project details, and payment information be securely stored and protected from unauthorized access. In the microservices design, only the Escrow & Payment Service and User & Auth Service, together with the encrypted Data Store, handle sensitive financial and identity data directly, each behind its own required/provided interface (EscrowAPI, UserAuth API, DB API). This narrows the attack surface: services like Project & Proposal or Notification never need direct access to payment or credential details, so a compromise of one service does not automatically expose sensitive data held by another.

Performance Benefit

Because each service is deployed and scaled independently behind the API Gateway, the Project & Proposal Service can be given more compute instances than the rest of the system precisely when browsing and search load spikes (for example, when several high-value projects are posted at once), directly supporting the 3-second response target in NFR-002 without requiring the entire application to be scaled as a single unit.

Component Diagram:

